

The Reversible Half and the Third Body: Pairs and Triads in Measurement Records and the Three-Body Problem

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Abstract

Time reversal is an involution: it splits every quantity into a part it keeps and a part it reverses. We determine what the reversed part carries in two settings, the classical records written by sequential quantum measurement and three equal masses under gravity. For a record chain C with stationary law π , time reversal $\tilde{C}$ and reversible half $S = \frac{1}{2}(C + \tilde{C})$, we prove that the entropy the record forgoes by being directed is a π -average of Jensen–Shannon divergences between its forward and time-reversed transition laws, and that $0 \leq h(S) - h(C) \leq \frac{1}{4}\text{EP}$, where EP is the entropy production, with equality to leading order as $C \rightarrow S$. The bound rests on the pointwise inequality $\text{JSD} \leq \frac{1}{8} \times (\text{Jeffreys divergence})$. For alternating sharp measurement of a singlet with $n \geq 3$ settings we solve the record exactly: the stationary record is reversible, and a record of length L carries total irreversibility $\frac{c}{2}(1 - d^{L-2}) \log_2 \frac{1+c}{1-c}$ bits, $c = \cos \frac{\pi}{n}$, $d = \cos \frac{2\pi}{n}$, whose limit $c \text{artanh}(c)$ nats is the Jeffreys divergence of the seed pair law from the stationary one; a separable preparation writes the identical record. A processed four-outcome instrument reconstructed from tomography writes its arrow continuously, at $\text{EP} = 0.00686044$ bits per use, with $h(S) - h(C)$ at 99.86% of $\text{EP}/4$. The rotating field of three-phase currents carries the same involution: time reversal transposes its lagged correlation matrix $K(\tau)$, whose symmetric half carries the magnitude $|I_+|^2 + |I_-|^2$ and whose antisymmetric half carries the signed swept area $\frac{9\pi}{4}(|I_+|^2 - |I_-|^2)$.

For three equal masses we prove that every collinear instant with one body at the mid-point is the triad $(-s, 0, s)$ about the centre of mass, that three bodies on one curve at thirds of the period carry no harmonic divisible by three, that the time-reversal symmetry of the figure-eight makes its angular momentum vanish harmonic by harmonic, as a balance of counter-rotating pairs, and that the equal-mass Lagrange triangle grows at exactly $\omega/\sqrt{2}$. The figure-eight has period $T = 6.3259140121$; it is a choreography to 1.3×10^{-8} from the published initial data and to 4.1×10^{-12} from refined data that close the orbit to 9.6×10^{-13} ; it holds the triad six times per period, at $kT/6$ to 5.5×10^{-9} , with each body at the zero twice; and it keeps its shape within 3.2×10^{-6} over 50 periods after a 10^{-6} kick, while the Lagrange triangle grows at 0.702194 against the exact $\omega/\sqrt{2} = 0.702331$. A fourth body of mass ε at distance D outside the Mardling–Aarseth boundary lends the triad a spin and takes it back every half outer orbit, deforming it by $\delta_{\max} \simeq \sqrt{3}\langle \Delta I \rangle \varepsilon D^{-3/2} = 2.84 \varepsilon D^{-3/2}$; the triad keeps its shape to 10^{-2} for $\varepsilon/D^3 \leq 10^{-4}$ and loses it for $\varepsilon/D^3 \geq 3.7 \times 10^{-4}$, and inside the boundary the fourth body is expelled. The proofs use convexity, Fourier series on the cube roots of unity and the symmetries of Newton’s equations; the orbits are integrated by an eighth-order Runge–Kutta method at tolerances 10^{-12} to 10^{-13} .

1 Introduction

A measurement record is written by an observer, and time reversal pairs every record with its reverse. Three bodies under gravity, which couples every pair of bodies with the same sign, carry a time reversal of their own. In both settings the involution splits every quantity into an

even half and an odd half. This paper determines what the odd half carries, how large it can be, and where it vanishes.

Records. Sequential use of a quantum instrument, an outcome map together with its conditional successor state, produces a classical record $X_0, X_1, \dots$ [15, 26]. Much is known about what such records certify (Bell nonlocality [1], contextuality [14], randomness) and what their erasure costs [16, 20, 4]. We ask whether the record *has a direction*: whether the statistics of the reversed sequence differ from those of the sequence itself, by how much, and what the difference costs in entropy. The natural measure is the entropy production of the record process [24, 10, 11], the rate at which forward and reversed records become distinguishable. When the record variables are a coarse description of the process that writes them, this statistical irreversibility bounds the physical dissipation from below and need not equal it [13, 22]; the corresponding statistic for continuous quantum measurement was studied by Dressel et al. [5]. The reversible chain $S = \frac{1}{2}(C + \tilde{C})$ is Fill’s additive reversibilization [7].

Three bodies. The prior results this paper builds on form a sequence (Table 1). Gascheau and Routh found when the rotating equilateral triangle of three masses is linearly stable [9, 23]. The figure-eight was found numerically by Moore [19]. Chenciner and Montgomery proved its existence and gave Simó’s initial data [2]: $x_1 = -x_2 = (0.97000436, -0.24308753)$, $x_3 = 0$, $\dot{x}_3 = -2\dot{x}_1 = -2\dot{x}_2 = (-0.93240737, -0.86473146)$, with unit masses, $G = 1$ and period 6.32591398. Simó studied its dynamics numerically [25]. Roberts proved its linear stability with rigorous numerical estimates: its nontrivial characteristic multipliers are distinct and lie on the unit circle [21]. Mardling and Aarseth gave the stability boundary of hierarchical triples used in Section 9 [17]. At $t = 0$ the figure-eight’s configuration is $(s, -s, 0)$, the triad itself.

year	authors	result
1843, 1874	Gascheau; Routh [9, 23]	the Lagrange triangle is linearly stable iff $27 \sum_{i < j} m_i m_j < (\sum_i m_i)^2$
1993	Moore [19]	the figure-eight found numerically
2000	Chenciner, Montgomery [2]	existence of the figure-eight; Simó’s initial data
2001	Mardling, Aarseth [17]	stability criterion for hierarchical triples
2002	Simó [25]	dynamical properties of the figure-eight
2007	Roberts [21]	linear stability of the figure-eight

Table 1: Prior results on three bodies used in this paper.

Companion papers. The singlet of Section 3 is the fourth state of the decomposition $\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$, whose triplet is the triad, and the identity $1 + u + u^2 = 0$ that removes the harmonics divisible by three in Theorem 1.4(b) is the identity that makes a probe qubit coupled to a maximally mixed spin-1 decohere completely at $\frac{1}{3}$ and $\frac{2}{3}$ of its revival period [27]. The restricted problem, where one of three masses vanishes, is treated in [28]: there each giant planet’s Hill sphere opens through two throats at L_1 and L_2 , a pair about the planet, and the triangular points are the Lagrange triangle of Section 6 with one vanishing mass, stable by the Gascheau–Routh criterion [9, 23]; the Trojans of the four giant planets and the fourth bodies that deplete them are treated there as well.

1.1 Main results

Let C be a column-stochastic matrix, $C_{ij} = \Pr(X_{t+1} = i \mid X_t = j)$, irreducible with stationary law π , stationary pair law $J_{ij} = \pi_j C_{ij}$ and time reversal $\tilde{C} = D_\pi C^\top D_\pi^{-1}$, $D_\pi = \text{diag}(\pi)$. Write

h for the entropy rate of a chain and $\text{EP}(C) = D(J \| J^\top)$ for the entropy production of the record (Theorem 1).

Theorem 1.1 (the reversible half). Assume $J_{ij} > 0 \iff J_{ji} > 0$, and write $C = S + A$ with $S = \frac{1}{2}(C + \tilde{C})$ and $A = \frac{1}{2}(C - \tilde{C})$. Then:

1. S is column-stochastic with stationary law π and satisfies detailed balance; A has zero column sums and $AD_\pi = \frac{1}{2}(J - J^\top)$ is antisymmetric.
2. $h(S) - h(C) = \sum_j \pi_j \text{JSD}(C_{\cdot j}, \tilde{C}_{\cdot j}) \geq 0$, where JSD is the Jensen–Shannon divergence.
3. $\text{EP}(C) = \sum_j \pi_j D(C_{\cdot j} \| \tilde{C}_{\cdot j}) = \sum_j \pi_j D(\tilde{C}_{\cdot j} \| C_{\cdot j})$, and

$$0 \leq h(S) - h(C) \leq \frac{1}{4} \text{EP}(C),$$

with equality in either inequality iff $A = 0$; as $A \rightarrow 0$ at fixed S , $h(S) - h(C) = \frac{1}{4} \text{EP}(C) (1 + O(\|A\|^2))$.

Theorem 1.2 (the arrow of the singlet walk). Let $n \geq 3$, $\vartheta = \pi/n$, $c = \cos \vartheta$, $d = \cos 2\vartheta$ and $O(\theta) = \cos \theta \sigma_z + \sin \theta \sigma_x$. Starting from the singlet $|\Psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$, measure $O(k\vartheta)$ sharply at step k on Alice for even k and on Bob for odd k , recording $X_k \in \{\pm 1\}$. Then $\Pr(X_0 = a) = \frac{1}{2}$, $\Pr(X_1 = b | X_0 = a) = \frac{1-abc}{2}$ and $\Pr(X_k = x | X_{<k}) = \frac{1+xX_{k-2}d}{2}$ for $k \geq 2$, and:

- (a) the stationary record, with (X_0, X_1) uniform and the same distance-two transitions, is reversible: every stationary block law is invariant under reversal;
- (b) for every record length $L \geq 2$ and every sequence, $\log \frac{P_{\rightarrow}(x_0, \dots, x_{L-1})}{P_{\rightarrow}(x_{L-1}, \dots, x_0)} = \log \frac{1-x_0x_1c}{1-x_{L-1}x_{L-2}c}$, and the total irreversibility of the record is

$$\mathcal{D}_L := D(P_{\rightarrow}^{(L)} \| P_{\leftarrow}^{(L)}) = \frac{c}{2}(1 - d^{L-2}) \log_2 \frac{1+c}{1-c} \text{ bits} \xrightarrow{L \rightarrow \infty} \frac{c \operatorname{artanh}(c)}{\ln 2} \text{ bits};$$

- (c) the limit equals the Jeffreys divergence between the law $\frac{1}{4}(1 - x_0x_1c)$ of (X_0, X_1) and the uniform law of an adjacent pair in the stationary record;
- (d) the separable state $\frac{1}{2}(|0\rangle\langle 0| \otimes |1\rangle\langle 1| + |1\rangle\langle 1| \otimes |0\rangle\langle 0|)$ writes the identical record law.

Theorem 1.3 (the rotating field). Let three coils at angles $0, \pm 120^\circ$ carry the currents $i_k(t) = \operatorname{Re}(I_k e^{i\omega t})$, $k \in \{a, b, c\}$, and let $B(t) = i_a(t) + u i_b(t) + u^{-1} i_c(t)$ be their field, $u = e^{2\pi i/3}$. With $I_\pm = \frac{1}{3}(I_a + u^{\pm 1} I_b + u^{\mp 1} I_c)$ and $I_0 = \frac{1}{3}(I_a + I_b + I_c)$,

$$B(t) = \alpha_+ e^{i\omega t} + \alpha_- e^{-i\omega t}, \quad \alpha_+ = \frac{3}{2} I_+, \quad \alpha_- = \frac{3}{2} \overline{I_-}.$$

1. The zero-sequence part I_0 does not contribute. Reversing the phase order ($b \leftrightarrow c$) exchanges I_+ and I_- , and reversing time ($t \mapsto -t$) exchanges the magnitudes of the forward and backward components.
2. The tip of B traces an ellipse with semi-axes $|\alpha_+| + |\alpha_-|$ and $||\alpha_+| - |\alpha_-||$, whose signed area per period is $\Phi = \pi(|\alpha_+|^2 - |\alpha_-|^2) = \frac{9\pi}{4}(|I_+|^2 - |I_-|^2)$.
3. Let $K(\tau) = \langle B(t+\tau)B(t)^\top \rangle$ be the lag- τ correlation matrix, averaged over a period, with $B = B_x + iB_y$ read as the column vector $(B_x, B_y)^\top$. Time reversal maps $K(\tau)$ to $K(\tau)^\top$, and

$$\operatorname{tr} K(\tau) = (|\alpha_+|^2 + |\alpha_-|^2) \cos \omega \tau, \quad K_{yx}(\tau) - K_{xy}(\tau) = \frac{\Phi}{\pi} \sin \omega \tau.$$

4. The following are equivalent: $K(\tau)$ is symmetric for every τ ; $\Phi = 0$; $|I_+| = |I_-|$; $B(t_0 + t) = B(t_0 - t)$ for some t_0 and all t .

Theorem 1.4 (three equal masses). In the planar Newtonian problem with $G = 1$ and three unit masses, with positions $x_i \in \mathbb{R}^2$ or $z_i \in \mathbb{C}$ and angular momentum $\mathcal{L} = \sum_i x_i \times \dot{x}_i$:

- (a) if the centre of mass is at the origin and at some instant body m lies at the midpoint of the other two, a and b , then $x_m = 0$ and $x_a = -x_b$;
- (b) if the centre of mass is at rest at the origin and $z_i(t) = q(t + (i-1)\nu T/3)$ for a T -periodic curve $q(t) = \sum_k \hat{q}_k e^{2\pi i k t/T}$ and a fixed $\nu \in \{1, -1\}$, then $\hat{q}_k = 0$ whenever $3 \nmid k$, and $\mathcal{L} = (6\pi/T) \sum_k k |\hat{q}_k|^2$;
- (c) for initial data with $x_3 = 0$, $x_1 = -x_2$ and $\dot{x}_1 = \dot{x}_2$, the solution satisfies $x_3(-t) = -x_3(t)$ and $x_1(-t) = -x_2(t)$; if it is a choreography as in (b), then $|\hat{q}_{-k}| = |\hat{q}_k|$ for all k , and $\mathcal{L} = 0$ harmonic by harmonic;
- (d) the equilateral triangle rotating rigidly at angular velocity ω is a relative equilibrium whose linearisation has the exponents $\omega(\pm 1/\sqrt{2} \pm i)$ and otherwise only 0 and $\pm i\omega$; its growth rate is $\max \operatorname{Re} \lambda = \omega/\sqrt{2}$.

Result 1.5 (the figure-eight and the triangle). *Integrated from the published initial data, the figure-eight has period $T = 6.3259140121$ and is a choreography, $x_{i+1}(t) = x_i(t - T/3)$, to 1.3×10^{-8} ; refined initial data close the orbit to 9.6×10^{-13} , with $T = 6.325914009781$, and give the choreography to 4.1×10^{-12} . The triangle of the bodies degenerates exactly six times per period, at $kT/6$ to 5.5×10^{-9} , each time into the triad $(-s, 0, s)$, with each body at the zero twice; the harmonics divisible by three are at most 7.2×10^{-13} of the largest. After a kick of 10^{-6} the eight keeps its shape within 3.2×10^{-6} over 50 periods, and within 5.3×10^{-6} after each of eight random kicks of the same size, while its equal-time distance from the unperturbed orbit grows linearly, by phase drift, to at most 1.68×10^{-3} . The Lagrange triangle with the same period leaves after every one of these kicks: after the kick to one body it grows at the rate 0.702194, against $\omega/\sqrt{2} = 0.702331$, and after the random kicks at 0.63197–0.70224.*

Result 1.6 (a fourth body). *Let a fourth body of mass $\varepsilon \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ start on a circular orbit at distance $D \in \{3, 5, 10, 20\}$ about the figure-eight. Every case with tidal parameter $\varepsilon/D^3 \leq 10^{-4}$ keeps the triad's shape to $\delta_{\max} < 10^{-2}$ over 100 periods, and every case with $\varepsilon/D^3 \geq 3.7 \times 10^{-4}$ does not; at fixed D the deformation is linear in ε to within a factor 1.244. To leading order it is set by the spin the fourth body lends and takes back, $\delta_{\max} \simeq 2 \max |\mathcal{L}_{\text{in}}| \simeq \sqrt{3} \langle \Delta I \rangle \varepsilon D^{-3/2} = 2.8438 \varepsilon D^{-3/2}$, and it is reversible: the triad returns to its unperturbed shape every half outer orbit. Outside the Mardling–Aarseth distance $D_{\text{MA}} = 4.06$ –4.11 the 3 + 1 pair stays bound; inside it, at $D = 3$, the fourth body is expelled in all four cases. Energy is conserved to 1.1×10^{-9} in every surviving case.*

1.2 Methods of proof

Theorem 1.1 reduces, column by column, to a comparison of two divergences. With $t = p_i/q_i$, the Jensen–Shannon and Jeffreys divergences are generated by $\varphi(t) = \frac{t}{2} \ln t - \frac{1+t}{2} \ln \frac{1+t}{2}$ and $\gamma(t) = (t-1) \ln t$, which vanish to first order at $t = 1$, and $\varphi'' \leq \gamma''/8$ because $4t \leq (1+t)^2$; Taylor's formula with integral remainder turns this into $\text{JSD} \leq \mathcal{J}/8$, and the Jeffreys divergences of the columns sum to 2 EP. For Theorem 1.2 the singlet's projector identity makes the state after the first outcome a product state, so the record consists of two independent distance-two Markov chains seeded by one correlated pair, and the log-likelihood ratio of a record and its reverse telescopes to the seed factor alone. Theorem 1.3 is Fortescue's decomposition into symmetrical components [8], read through the lagged correlation matrix. Theorem 1.4 uses the identity $1+u+u^2=0$ for the Fourier coefficients of a choreography, the invariance of Newton's equations under $(t, x_1, x_2, x_3) \mapsto (-t, -x_2, -x_1, -x_3)$, and, for the Lagrange triangle, a linearisation in complex rotating coordinates in which the mode $\chi_j = u^{2j}b$ decouples. Results 1.5 and 1.6 are computed by integrating Newton's equations with the eighth-order Dormand–Prince method with dense output [12], and the leading-order law of Result 1.6 follows from the quadrupole tidal torque averaged over the inner period.

1.3 Organisation and notation

Section 2 proves Theorem 1.1, Section 3 proves Theorem 1.2, and Section 4 applies both to a processed four-outcome instrument. Section 5 proves Theorem 1.3 and Section 6 proves Theorem 1.4. Sections 7 and 8 give Result 1.5 and Section 9 gives Result 1.6. Section 10 sets the six systems side by side as instances of one involution. Logarithms are to base 2 (bits) or natural (nats), as stated. H is the Shannon entropy, $D(P\|Q)$ the Kullback–Leibler divergence, $\text{JSD}(P, Q) = H(\frac{P+Q}{2}) - \frac{1}{2}H(P) - \frac{1}{2}H(Q)$ the Jensen–Shannon divergence and $\mathcal{J}(P, Q) = D(P\|Q) + D(Q\|P)$ the Jeffreys divergence; $u = e^{2\pi i/3}$ throughout. Every number is computed from the stated closed forms, from the stated transition matrix, or by numerical integration of Newton’s equations, by the method given where the number appears, in IEEE double precision unless stated otherwise.

2 Records, reversal and the reversible half

Let C , π , D_π and J be as in Section 1.1, and write $\Pi = \pi \mathbf{1}^\top$. The time-reversed chain $\tilde{C} = D_\pi C^\top D_\pi^{-1}$, i.e. $\tilde{C}_{ij} = J_{ji}/\pi_j$, has the same π and the same entropy rate $h(C) = -\sum_{ij} J_{ij} \log C_{ij}$. Indeed $\sum_i J_{ij} = \pi_j$ (columns of C sum to one) and $\sum_j J_{ij} = (C\pi)_i = \pi_i$ (stationarity), so $\sum_i \tilde{C}_{ij} = \pi_j^{-1} \sum_i J_{ji} = 1$ and $\sum_j \tilde{C}_{ij} \pi_j = \sum_j J_{ji} = \pi_i$; and $h(C) = -\sum_{ij} J_{ij} \log J_{ij} + \sum_{ij} J_{ij} \log \pi_j = H(J) - H(\pi)$, while the pair law of $\tilde{C}$ is $\pi_j \tilde{C}_{ij} = J_{ji}$, so $h(\tilde{C}) = H(J^\top) - H(\pi) = h(C)$.

Definition 1. Assume $J_{ij} > 0 \iff J_{ji} > 0$ (otherwise the quantity below is $+\infty$). The entropy production of the record is $\text{EP}(C) = \sum_{ij} J_{ij} \log \frac{J_{ij}}{J_{ji}} = D(J\|J^\top)$; it equals $\lim_{L \rightarrow \infty} \frac{1}{L} D(P_{\rightarrow}^{(L)}\|P_{\leftarrow}^{(L)})$ for stationary records of length L . For a cycle $i_1 \rightarrow \dots \rightarrow i_m \rightarrow i_1$ the affinity is $\log \prod_k C_{i_{k+1}i_k} / \prod_k C_{i_k i_{k+1}}$.

The limit in the Definition holds in the exact form $D(P_{\rightarrow}^{(L)}\|P_{\leftarrow}^{(L)}) = (L-1) \text{EP}$. For a stationary record, $P_{\rightarrow}(x_0, \dots, x_{L-1}) = \pi_{x_0} \prod_{t=0}^{L-2} C_{x_{t+1}x_t}$ and $P_{\leftarrow}(x_0, \dots, x_{L-1}) = P_{\rightarrow}(x_{L-1}, \dots, x_0) = \pi_{x_{L-1}} \prod_{t=0}^{L-2} C_{x_t x_{t+1}}$, so

$$\log \frac{P_{\rightarrow}}{P_{\leftarrow}} = \log \frac{\pi_{x_0}}{\pi_{x_{L-1}}} + \sum_{t=0}^{L-2} \log \frac{C_{x_{t+1}x_t}}{C_{x_t x_{t+1}}}.$$

Under $P_{\rightarrow}$ both x_0 and x_{L-1} have law π , so the first term has mean zero, and each pair (x_t, x_{t+1}) has law J ; since $\log \frac{C_{ij}}{C_{ji}} = \log \frac{J_{ij}}{J_{ji}} + \log \frac{\pi_i}{\pi_j}$ and $\sum_{ij} J_{ij} \log \frac{\pi_i}{\pi_j} = \sum_i \pi_i \log \pi_i - \sum_j \pi_j \log \pi_j = 0$, the mean of the sum is $(L-1) \text{EP}$ [10, 11]. $\text{EP} \geq 0$, with equality iff $J = J^\top$ (detailed balance), by Gibbs’ inequality for the two pair laws J and $J^\top$. Detailed balance holds iff every cycle affinity vanishes (Kolmogorov’s criterion): if $J = J^\top$ then $C_{ij}/C_{ji} = \pi_i/\pi_j$ and the product round any cycle telescopes to 1; conversely, if every cycle affinity vanishes, fix a state o and let μ_i be the product of $C_{i_{k+1}i_k}/C_{i_k i_{k+1}}$ along a path of allowed transitions from o to i (independent of the path, because two paths close into a cycle); then $\mu_j C_{ij} = \mu_i C_{ji}$ for all i, j , summing over j gives $C\mu = \mu$, so $\mu \propto \pi$ by uniqueness of the stationary law, and $J = J^\top$.

We now prove Theorem 1.1.

Proof of Theorem 1.1. (1) Since $\tilde{C}D_\pi = D_\pi C^\top = (CD_\pi)^\top$ and $CD_\pi = J$, $SD_\pi = \frac{1}{2}(J + J^\top)$ is symmetric, which is detailed balance for S , and $AD_\pi = \frac{1}{2}(J - J^\top)$ is antisymmetric; column sums and π follow from those of C and $\tilde{C}$ shown above. (2) Since π is stationary for S , $h(S) = \sum_j \pi_j H(S_{\cdot j}) = \sum_j \pi_j H(\frac{1}{2}C_{\cdot j} + \frac{1}{2}\tilde{C}_{\cdot j})$, and $\text{JSD}(P, Q) = H(\frac{P+Q}{2}) - \frac{1}{2}H(P) - \frac{1}{2}H(Q)$, which is nonnegative by concavity of H ; sum with weights π_j and use $\sum_j \pi_j H(C_{\cdot j}) = h(C)$

and $\sum_j \pi_j H(\tilde{C}_{\cdot j}) = h(\tilde{C}) = h(C)$. (3) $\sum_j \pi_j D(C_{\cdot j} \| \tilde{C}_{\cdot j}) = \sum_{ij} J_{ij} \log \frac{J_{ij}/\pi_j}{J_{ji}/\pi_i} = D(J \| J^\top)$ and likewise $\sum_j \pi_j D(\tilde{C}_{\cdot j} \| C_{\cdot j}) = D(J^\top \| J)$; the two are equal by relabelling $i \leftrightarrow j$. So $\sum_j \pi_j \mathcal{J}(C_{\cdot j}, \tilde{C}_{\cdot j}) = 2 \text{EP}$, where $\mathcal{J}$ is the Jeffreys divergence. For $t > 0$ put $\varphi(t) = \frac{t}{2} \ln t - \frac{1+t}{2} \ln \frac{1+t}{2}$ and $\gamma(t) = (t-1) \ln t$; then $\text{JSD}(P, Q) = \sum_i q_i \varphi(p_i/q_i)$ and $\mathcal{J}(P, Q) = \sum_i q_i \gamma(p_i/q_i)$. Both functions vanish with their first derivatives at $t = 1$, and

$$\varphi''(t) = \frac{1}{2t(1+t)} \leq \frac{1+t}{8t^2} = \frac{1}{8} \gamma''(t) \quad (4t \leq (1+t)^2),$$

with equality only at $t = 1$. By Taylor's formula with integral remainder, $\varphi(t) = \int_1^t (t-r) \varphi''(r) dr$ and $\gamma(t) = \int_1^t (t-r) \gamma''(r) dr$; for $t > 1$ the weight $t-r$ is nonnegative on $[1, t]$, and for $t < 1$, $\int_1^t (t-r) g(r) dr = \int_t^1 (r-t) g(r) dr$ with $r-t \geq 0$; in both cases $\varphi'' \leq \gamma''/8$ integrates to $\varphi(t) \leq \gamma(t)/8$, with equality only at $t = 1$. Hence $\text{JSD} \leq \mathcal{J}/8$ columnwise (the φ, γ forms need $q_i > 0$; indices with $p_i = q_i = 0$ contribute nothing, and $p_i > 0 \iff q_i > 0$ by the standing assumption), and summing with weights π_j gives the bound. Equality, in either inequality, forces $C_{\cdot j} = \tilde{C}_{\cdot j}$ for every j , i.e. $A = 0$. For the leading order, put $C_{\cdot j} = v + a$, $\tilde{C}_{\cdot j} = v - a$ with $v = S_{\cdot j}$, $a = A_{\cdot j}$ (so $\sum_i a_i = 0$). Both divergences are unchanged under $P \leftrightarrow Q$, i.e. $a \mapsto -a$, so their expansions in a are even. From $D(v \pm a \| v) = \frac{1}{2} \sum_i a_i^2 / v_i + O(\|a\|^3)$, $\text{JSD} = \frac{1}{2} D(v+a \| v) + \frac{1}{2} D(v-a \| v) = \frac{1}{2} \sum_i a_i^2 / v_i + O(\|a\|^4)$, and $\mathcal{J} = \sum_i 2a_i \ln \frac{v_i+a_i}{v_i-a_i} = 4 \sum_i a_i^2 / v_i + O(\|a\|^4)$ (nats); the ratio is $\frac{1}{8}(1 + O(\|a\|^2))$, and summing with weights π_j and using $\sum_j \pi_j \mathcal{J}(C_{\cdot j}, \tilde{C}_{\cdot j}) = 2 \text{EP}$ gives the last claim. $\square$

The reversible half carries the randomness of the record; the odd half carries its direction, and the entropy the record forgoes by being directed is at most, and to leading order exactly, one quarter of what its direction produces. S is Fill's additive reversibilization [7]; the identity (2) and the bound (3) of Theorem 1.1 do not appear there.

3 Sharp measurement of a singlet: an arrow set by the seed

Let n, ϑ, c, d and $O(\theta)$ be as in Theorem 1.2, and start from the singlet $|\Psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$ [26]. Write $|a; \theta\rangle$ for the eigenvector of $O(\theta)$ with eigenvalue a ; its Bloch vector is the unit vector at angle θ in the zx -plane, so a sharp measurement of $O(\theta')$ on it gives x with probability $\frac{1}{2}(1 + xa \cos(\theta' - \theta))$. The singlet satisfies $(P_a(\theta) \otimes I) |\Psi^-\rangle \propto |a; \theta\rangle \otimes |-a; \theta\rangle$ for every θ , with $P_a(\theta) = \frac{1}{2}(I + aO(\theta))$, so the first outcome is uniform and leaves the product state $|a; 0\rangle \otimes |-a; 0\rangle$, and local sharp measurements preserve product form. At step $k \geq 1$ the measured qubit is in the state left by its own last measurement: $|-X_0; 0\rangle$ for Bob at $k = 1$, and $|X_{k-2}; (k-2)\vartheta\rangle$ for $k \geq 2$. Hence

$$\Pr(X_0 = a) = \frac{1}{2}, \quad \Pr(X_1 = b \mid X_0 = a) = \frac{1-abc}{2}, \quad \Pr(X_k = x \mid X_{<k}) = \frac{1+xX_{k-2}d}{2} \quad (k \geq 2),$$

which is the first assertion of Theorem 1.2. The adjacent correlator is $\rho_k = \mathbb{E}[X_k X_{k+1}] = -c d^k$: $\rho_0 = \sum_{a,b} ab \cdot \frac{1}{4}(1-abc) = -c$, and $\mathbb{E}[X_{k+1} \mid X_{\leq k}] = dX_{k-1}$ gives $\rho_k = d\rho_{k-1}$. After the first step the even and odd subsequences are independent symmetric Markov chains with flip probability $\frac{1-d}{2}$. Each has mean zero, and the pair (X_k, X_{k+1}) has mean-zero marginals and correlation $\rho_k \rightarrow 0$, so its law tends to the uniform law on $\{\pm 1\}^2$; the *stationary record* is the process with (X_0, X_1) uniform and the same distance-two transitions. We now prove parts (a)–(c) of Theorem 1.2.

Proof of Theorem 1.2(a)–(c). (a) In the stationary record the even and odd subsequences are independent stationary symmetric two-state chains with the same law, and each is reversible (a symmetric transition matrix with uniform stationary law satisfies detailed balance). Reversal

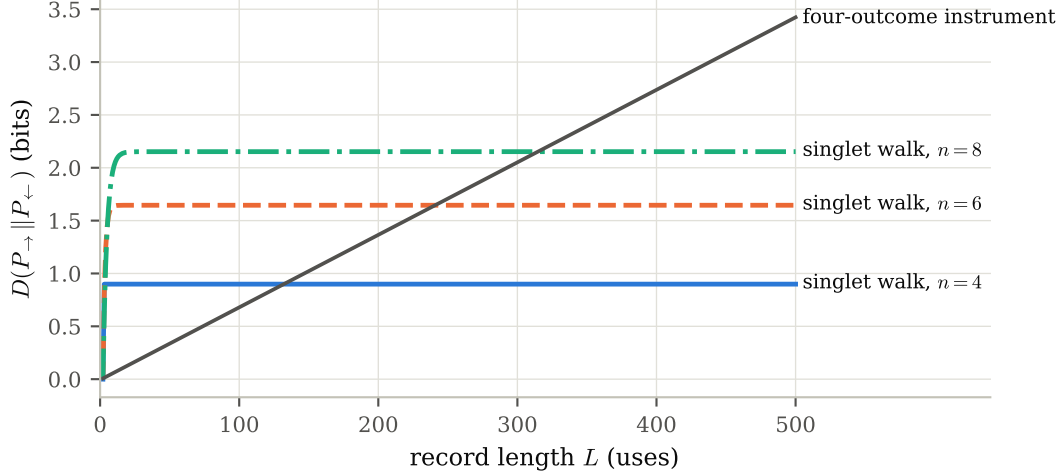


Figure 1: Total irreversibility $D(P_{\rightarrow} \| P_{\leftarrow})$ of a record of length L . Singlet walks ($n = 4, 6, 8$) saturate at $c \operatorname{artanh}(c) / \ln 2 = 0.899, 1.645, 2.152$ bits, the Jeffreys divergence of their seed pair law from the stationary one. The processed four-outcome instrument of Section 4 grows linearly, as $(L - 1)$ EP with EP = 0.00686044 bits per use, and overtakes the $n = 4$ total after 132 uses.

$k \mapsto L - 1 - k$ maps each subsequence, reversed, onto one of the two (itself if L is odd, the other if L is even); since they are independent, identically distributed and reversible, every block law is reversal-invariant. In particular X_{k-1}, X_k are independent and uniform and $\Pr(X_{k+1} = e \mid X_{k-1} = a) = \frac{1+aed}{2}$, so the triple law $\frac{1}{4} \cdot \frac{1+aed}{2}$ on $(X_{k-1}, X_k, X_{k+1}) = (a, b, e)$ is symmetric in (a, e) , and $\mathcal{D}_L/L \rightarrow 0$ by (b). (b) The factors $\frac{1+x_k x_{k-2} d}{2}$, $2 \leq k \leq L - 1$, range over all index pairs at distance 2, a set invariant under $k \mapsto L - 1 - k$; only the seed factor changes. Under $P_{\rightarrow}$, $x_0 x_1 = -1$ with probability $\frac{1+c}{2}$ and $x_{L-2} x_{L-1} = -1$ with probability $\frac{1-\rho_{L-2}}{2} = \frac{1+cd^{L-2}}{2}$; taking expectations of the log-ratio,

$$\mathcal{D}_L = \left[\frac{1+c}{2} - \frac{1+cd^{L-2}}{2} \right] \log(1+c) + \left[\frac{1-c}{2} - \frac{1-cd^{L-2}}{2} \right] \log(1-c) = \frac{c}{2} (1 - d^{L-2}) \log \frac{1+c}{1-c},$$

and $\frac{1}{2} \ln \frac{1+c}{1-c} = \operatorname{artanh}(c)$. (c) $\sum_{x_0, x_1} (\frac{1}{4}(1 - x_0 x_1 c) - \frac{1}{4}) \ln(1 - x_0 x_1 c) = \frac{c}{2} \ln \frac{1+c}{1-c} = c \operatorname{artanh}(c)$. $\square$

Remark 2 (the pair chain). The pair process (X_{k-1}, X_k) , viewed as a first-order chain on $\{\pm 1\}^2$, has transitions $(a, b) \rightarrow (b, e)$, half of whose reverses (those with $a \neq e$) are forbidden, so its entropy production in the sense of Theorem 1 is infinite. Reversibility in Theorem 1.2(a) is a statement about the law of the sequence, which is reversal-invariant.

Values. For $n = 4$, $d = 0$ and $\mathcal{D}_L = 0.899124$ bits for every $L \geq 3$; for $n = 6$, $\mathcal{D}_L = 0.822711, 1.234066, \dots, 1.645019$ for $L = 3, \dots, 14$, with limit 1.645421; for $n = 8$, $d < 0$ and $\mathcal{D}_L$ oscillates to its limit 0.396241. Direct summation over all 2^L sequences for $L \leq 14$ and $n = 3, 4, 5, 6, 8, 16$ agrees with the closed form to 4×10^{-13} , and the record law computed from the two-qubit density matrix itself, applying the projectors $P_{X_k}(k\vartheta)$ to the singlet in turn for $n = 3, 4, 5, 6, 8$ and all 2^8 sequences of length 8, agrees with the law of Theorem 1.2 to 6×10^{-17} . As $n \rightarrow \infty$, $1 - c = \frac{\pi^2}{2n^2} + O(n^{-4})$, so $c \operatorname{artanh}(c) = \ln \frac{2n}{\pi} + O(n^{-2} \ln n)$: the limit grows like $\log_2(2n/\pi)$ bits.

What sets the arrow. By (c) the arrow is a property of the seed: the first two outcomes are correlated ($\rho_0 = -c$), the stationary record has uncorrelated adjacent outcomes, and the total irreversibility is exactly the Jeffreys divergence between the two pair laws, realised gradually as the record relaxes at rate d per step. The arrow is independent of entanglement, which is

Theorem 1.2(d): the first measurement is of $O(0) = \sigma_z$ on Alice, and after its outcome the singlet is in exactly one of the product states $|0\rangle \otimes |1\rangle$, $|1\rangle \otimes |0\rangle$, with probabilities $\frac{1}{2}, \frac{1}{2}$, which is the state the separable preparation is in after the same outcome; every later step acts on these product states in the same way. The density-matrix computation above gives the two record laws equal to 6×10^{-17} . What the singlet adds is frame independence: its record law is unchanged by a common rotation of all settings, whereas a separable preparation reproduces it only for the settings it is aligned with. The singlet is the fourth state of $\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$, whose triplet $m = -1, 0, 1$ is the triad [27]; the walk writes its record from that fourth state.

4 A processed four-outcome instrument: an arrow written continuously

The second instrument is the processed four-outcome instrument of [26]: a Bell instrument reconstructed from tomography, with effects $E_a = \sum_r e_{ar} |v_{ar}\rangle \langle v_{ar}|$ and conditional output states $\varrho_a = \sum_s q_{as} |w_{as}\rangle \langle w_{as}|$ in spectral form, physically completed by the Kraus family $K_{asr} = \sqrt{e_{ar} q_{as}} |w_{as}\rangle \langle v_{ar}|$, with one nonpositive estimate repaired by the maximal positivity factor 0.708418699729814, and iterated in a common basis. Since $\sum_{s,r} K_{asr}^\dagger K_{asr} = E_a$ and $\sum_{s,r} K_{asr} \varrho K_{asr}^\dagger = \text{Tr}(E_a \varrho) \varrho_a$, the instrument is measure-and-prepare: after summing over the unrecorded labels r, s the post-measurement state depends only on the recorded outcome a , so the record is a first-order Markov chain, with $C_{ij} = \text{Tr}(E_i \varrho_j)$. Its transition law is

$$C = \begin{pmatrix} 0.43696297 & 0.19810609 & 0.20644079 & 0.14060033 \\ 0.16017410 & 0.46165671 & 0.13951861 & 0.22362294 \\ 0.24443666 & 0.11892971 & 0.46061556 & 0.17659060 \\ 0.15842628 & 0.22130750 & 0.19342504 & 0.45918613 \end{pmatrix}.$$

We use this eight-digit matrix as given (two of its columns sum to 1.00000001; renormalising them changes the quantities below by at most one unit in the last digit shown). Everything in this section is a property of this processed instrument. Its stationary law, the normalised eigenvector for eigenvalue 1, and its spectrum are

$$\begin{aligned} \pi &= (0.24327934, 0.24563854, 0.24991431, 0.26116781), \\ \text{spec } C &= \{1, 0.35301359, 0.27498875, 0.19041903\}; \end{aligned}$$

the stationary entropy $H(X_t) = H(\pi) = 1.9994586$ bits, the entropy rate $h = H(J) - H(\pi) = 1.8413629$ bits per use and $I(X_t; X_{t+1}) = H(\pi) - h = 0.15809572$ bits; the minimal record cost $k_B T_0 \ln 2 \cdot h = 1.2763355 k_B T_0$ per use, for a bath at temperature T_0 , with the previous record as side information [4]; and the centred Poisson potential $\kappa = (0.03023949, -0.02314357, 0.00046184, -0.00684277)$, the unique solution of $(I - C^\top) \kappa = g - h \mathbf{1}$ with $\pi^\top \kappa = 0$, where $g_j = -\sum_i C_{ij} \log_2 C_{ij}$ is the expected surprisal of a use made from record j . (The equation is solvable because $\pi^\top (g - h \mathbf{1}) = 0$ and the range of $I - C^\top$ is the orthogonal complement of the kernel of $I - C$, which is spanned by π ; its solutions differ by multiples of 1.) These agree with the values given in [26] to the rounding of C .

The arrow. Detailed balance fails ($\max |J - J^\top| = 0.00970$) and

$$\text{EP} = 0.00686044 \text{ bits per use} = 0.373\% \text{ of } h.$$

The four three-cycle affinities are $\mathcal{A}_{012} = -0.7807$ ($0 \rightarrow 1 \rightarrow 2 \rightarrow 0$), $\mathcal{A}_{013} = -0.4939$ ($0 \rightarrow 1 \rightarrow 3 \rightarrow 0$), $\mathcal{A}_{023} = +0.2029$ ($0 \rightarrow 2 \rightarrow 3 \rightarrow 0$) and $\mathcal{A}_{123} = -0.0840$ ($1 \rightarrow 2 \rightarrow 3 \rightarrow 1$) bits. Only three are independent (the complete graph on four states has three independent

cycles): both $\mathcal{A}_{012} + \mathcal{A}_{023}$ and $\mathcal{A}_{013} + \mathcal{A}_{123}$ equal the four-cycle affinity -0.5778 . The record circulates round $0 \rightarrow 2 \rightarrow 1 \rightarrow 0$ with odds $2^{0.7807} = 1.718 : 1$ over the reverse. The reversible half S has $h(S) = 1.8430756$, so $h(S) - h(C) = \sum_j \pi_j \text{JSD} = 0.00171268$, below the bound $\text{EP}/4 = 0.00171511$ of Theorem 1.1 (ratio 0.99858), and the four columnwise ratios $\text{JSD}/\mathcal{J}$ lie between 0.12479 and 0.12498, against the ceiling $\frac{1}{8}$. The odd half is small, $\|A\|_F = 0.0418$ against $\|S - \Pi\|_F = 0.4882$. For a stationary record of length L the accumulated irreversibility is exactly $(L-1) \text{EP}$ (Section 2; the stationary boundary terms cancel), which exceeds the total 0.899124 bits of the $n = 4$ singlet walk once $L - 1 \geq 0.899124/0.00686044 = 131.06$, i.e. after 132 uses (Fig. 1). The constant $\frac{1}{4}$ is sharp (Theorem 1.1(3)): over 2×10^4 random chains with 3 to 6 states, whose columns are drawn from symmetric Dirichlet laws with parameter uniform on $[0.2, 5]$, the largest ratio $(h(S) - h(C))/(\text{EP}/4)$ is $1 - 1.1 \times 10^{-10}$ (the 1820 chains with a columnwise ratio $\text{JSD}/\mathcal{J}$ above 0.1249 were evaluated in 60-significant-digit arithmetic).

Fluctuations of the record cost. Let $f_t = F_{X_{t+1}X_t}$ with $F_{ij} = -\log_2 C_{ij}$ be the per-use surprisal, whose stationary mean is h and whose conditional mean given $X_t = j$ is g_j . Its stationary variance is $\text{Var}(f_0) = \sum_{ij} J_{ij} (F_{ij} - h)^2 = 0.46159536$ bits². For $t \geq 1$, $\mathbb{E}[f_t - h \mid X_1 = i] = [(C^\top)^{t-1}(g - h\mathbf{1})]_i$, and $\sum_{r \geq 0} (C^\top)^r (g - h\mathbf{1}) = \kappa$ (the series converges geometrically, at the rate 0.353 of the second eigenvalue, is centred because $\pi^\top C^\top = \pi^\top$, and solves the Poisson equation), so the lagged covariances sum in closed form to

$$\sum_{t \geq 1} \text{Cov}(f_0, f_t) = \sum_{ij} J_{ij} (F_{ij} - h) \kappa_i = 0.00044146 \text{ bits}^2,$$

and the Markov central limit theorem [18] gives

$$\sigma^2 = \text{Var}(f_0) + 2 \sum_{t \geq 1} \text{Cov}(f_0, f_t) = 0.46247828 \text{ bits}^2 \text{ per use.}$$

The empirical rate after N uses therefore lies in $h \pm 1.96\sigma/\sqrt{N} = h \pm 1.333/\sqrt{N}$ bits with 95% probability (to leading order in N). Beside this there is a deterministic bias: for a record started at x_0 , $g - h\mathbf{1} = \kappa - C^\top \kappa$ gives $\mathbb{E}[g_{X_t}] - h = \mathbb{E}[\kappa_{X_t}] - \mathbb{E}[\kappa_{X_{t+1}}]$, so $\mathbb{E} \sum_{t < N} (f_t - h) = \kappa_{x_0} - \mathbb{E}[\kappa_{X_N}]$, and the bias of the rate is at most $2 \max_j |\kappa_j|/N = 0.0605/N$ bits [26]. Under an accounting that charges each record its surprisal, the cost of N uses has mean $1.2763355 N k_B T_0$ and standard deviation $0.680 \ln 2 \sqrt{N} = 0.471 \sqrt{N} k_B T_0$; Landauer's principle bounds the heat of erasing the record from below by the mean [16, 20, 4]. The total-variation distance of the law of X_t to π , maximised over the initial record (the largest half column-sum of $|C^t - \Pi|$), is 0.216, 0.069, 3.2×10^{-3} , 1.8×10^{-5} after $t = 1, 2, 5, 10$ uses (spectral gap $1 - 0.353 = 0.647$).

5 The same involution in a rotating field

Three coils at angles $0, \pm 120^\circ$ carrying alternating currents produce, in complex notation, the rotating field $B(t)$ of Theorem 1.3 [6, 30]. We identify $B = B_x + iB_y$ with the column vector $(B_x, B_y)^\top$, and now prove Theorem 1.3.

Proof of Theorem 1.3. $i_k = \frac{1}{2}(I_k e^{i\omega t} + \bar{I}_k e^{-i\omega t})$, $I_a + uI_b + u^{-1}I_c = 3I_+$ and $\bar{I}_a + u\bar{I}_b + u^{-1}\bar{I}_c = \bar{I}_a + u^{-1}\bar{I}_b + u\bar{I}_c = 3\bar{I}_-$, which gives the form of B . (1) Since $1 + u + u^{-1} = 0$, adding a common I_0 to all three currents changes nothing. Exchanging I_b and I_c exchanges the definitions of I_+ and I_- . $B(-t) = \alpha_+ e^{-i\omega t} + \alpha_- e^{i\omega t}$, whose forward and backward amplitudes are $|\alpha_-|$ and $|\alpha_+|$. (2) $\bar{B}\dot{B} = i\omega(|\alpha_+|^2 - |\alpha_-|^2 + z - \bar{z})$ with $z = \bar{\alpha}_- \alpha_+ e^{2i\omega t}$, and $i\omega(z - \bar{z})$ is real, so $B_x \dot{B}_y - B_y \dot{B}_x = \text{Im}(\bar{B}\dot{B}) = \omega(|\alpha_+|^2 - |\alpha_-|^2)$ is constant, and over a period $2\pi/\omega$ the signed area $\frac{1}{2} \oint (B_x dB_y - B_y dB_x) = \pi(|\alpha_+|^2 - |\alpha_-|^2)$. The sum of two counter-rotating circular

motions is an ellipse with semi-axes $|\alpha_+| + |\alpha_-|$ and $||\alpha_+| - |\alpha_-||$. (3) For the reversed field $B_r(t) = B(-t)$, the substitution $t' = -t - \tau$, under which the period average is invariant, gives $\langle B_r(t + \tau) B_r(t)^\top \rangle = \langle B(t') B(t' + \tau)^\top \rangle = K(\tau)^\top$. Next, $\langle \bar{B}(t) B(t + \tau) \rangle = |\alpha_+|^2 e^{i\omega\tau} + |\alpha_-|^2 e^{-i\omega\tau}$, because the cross terms carry $e^{\pm 2i\omega t}$ and average to zero. Its real part is $\langle B_x(t) B_x(t + \tau) + B_y(t) B_y(t + \tau) \rangle = \text{tr } K(\tau)$ and its imaginary part is $\langle B_x(t) B_y(t + \tau) - B_y(t) B_x(t + \tau) \rangle = K_{yx}(\tau) - K_{xy}(\tau)$. (4) The antisymmetric part of a 2×2 matrix is fixed by $K_{yx} - K_{xy}$, which by (3) vanishes for every τ iff $\Phi = 0$, i.e. iff $|I_+| = |I_-|$. If $|\alpha_+| = |\alpha_-| = r$ and $\alpha_\pm = r e^{i\theta_\pm}$, then $B(t) = 2r e^{i(\theta_+ + \theta_-)/2} \cos(\omega t + \frac{\theta_+ - \theta_-}{2})$, a linear oscillation symmetric about $t_0 = -(\theta_+ - \theta_-)/2\omega$. Conversely, if $B(t_0 + t) = B(t_0 - t)$, then $B_x \dot{B}_y - B_y \dot{B}_x$ changes sign under $t \mapsto 2t_0 - t$; being constant, it is zero, and $\Phi = 0$. $\square$

A single coil gives $|I_+| = |I_-| = \frac{1}{3}|I_a|$: its field pulsates with zero swept area. A balanced set in one order is pure forward rotation (area $\frac{9\pi}{4}|I|^2 = 7.0686$ for $|I| = 1$), in the other pure backward rotation. For the unbalanced example $(1, 0.8\angle -120^\circ, 1.1\angle 118^\circ)$ the forward and backward amplitudes $|\alpha_+|$ and $|\alpha_-|$ are 1.4498 and 0.1203 and the swept area is $\Phi = +6.5578$. The decomposition is Fortescue's method of symmetrical components [8, 3].

The shared structure. Theorems 1.1 and 1.3 are two instances of one structure. A stationary process W_t with values in $\mathbb{R}^m$ has lagged correlation matrices $K(\tau) = \mathbb{E}[W_{t+\tau} W_t^\top]$, and time reversal acts on them by transposition, $K(\tau) \mapsto K(\tau)^\top = K(-\tau)$. For the record, $W_t = e_{X_t}$ is the indicator vector of the outcome and $K(1) = J$, so time reversal is $J \mapsto J^\top$, that is $C \mapsto \bar{C}$; for the field, $W_t = B(t)$. In both, the symmetric half carries magnitude: S , the reversible chain with the same π and the larger entropy rate; and $\text{tr } K(0) = |\alpha_+|^2 + |\alpha_-|^2$, the mean-square field. In both, the antisymmetric half carries a signed circulation: the cycle currents $J - J^\top = 2AD_\pi$, which vanish exactly when every cycle affinity does; and the swept area Φ , through $K_{yx} - K_{xy} = (\Phi/\pi) \sin \omega\tau$. In both, the antisymmetric half vanishes exactly on the reversible set: $J = J^\top$ (detailed balance, every affinity zero), and $|I_+| = |I_-|$ (the field is a linear oscillation, reversible about t_0). The two differ in how distance from the reversible set is measured: EP is a divergence of J from $J^\top$ and is nonnegative, whereas Φ is linear in the antisymmetric half and carries its sign, as the cycle currents do.

6 Three bodies: exact statements

Throughout Sections 6 to 9 the model is the planar Newtonian problem in dimensionless units, $G = 1$, with unit masses for the triad; positions are $x_i \in \mathbb{R}^2$, or $z_i \in \mathbb{C}$ in complex notation, and $\mathcal{L} = \sum_i m_i x_i \times \dot{x}_i$ is the angular momentum. We prove the four parts of Theorem 1.4 in turn. Part (a) is a statement about the centre of mass.

Proof of Theorem 1.4(a). $x_a + x_b = 2x_m$ and $x_a + x_b + x_m = 0$ give $3x_m = 0$, and then $x_a = -x_b$. $\square$

The middle body is the zero of the pair $\pm s$, at the pair's centre [29]. Part (b) is a statement about Fourier coefficients.

Proof of Theorem 1.4(b). $\sum_i z_i(t) = \sum_k \hat{q}_k e^{2\pi i k t/T} (1 + u^{\nu k} + u^{2\nu k})$ with $u = e^{2\pi i/3}$. The bracket is 3 if $3 \mid k$ and 0 otherwise, so $\sum_i z_i \equiv 0$ forces $\hat{q}_k = 0$ for $3 \nmid k$, by uniqueness of Fourier coefficients. $\text{Im}(\bar{z}\dot{z})$ is a unit mass's angular momentum. Since $\dot{q} = \sum_k (2\pi i k/T) \hat{q}_k e^{2\pi i k t/T}$, Parseval's identity gives the period average $\langle \bar{q}\dot{q} \rangle = (2\pi i/T) \sum_k k |\hat{q}_k|^2$, so each body, whose path is a time shift of q , has average angular momentum $(2\pi/T) \sum_k k |\hat{q}_k|^2$. The total $\mathcal{L}$ is constant, hence equal to its average, three times this. $\square$

The phases $1, u, \bar{u}$ are the triad of the circle: the pair $u, \bar{u}$ sums to -1 and balances the third. The centre of mass sees only the harmonics in which this balance fails, and these must vanish. The same identity $1 + u + u^2 = 0$ removes the common mode I_0 from the field of Theorem 1.3, which is built from the phases $1, u, u^{-1}$ of the three coils. Part (c) rests on a symmetry of Newton's equations.

Proof of Theorem 1.4(c). For equal masses, Newton's equations $\ddot{x}_i = \sum_{j \neq i} (x_j - x_i)/|x_j - x_i|^3$ are invariant under $(t, x_1, x_2, x_3) \mapsto (-t, -x_2, -x_1, -x_3)$. Indeed, write $F_i(x) = \sum_{j \neq i} (x_j - x_i)/|x_j - x_i|^3$ and $y_1(t) = -x_2(-t)$, $y_2(t) = -x_1(-t)$, $y_3(t) = -x_3(-t)$. Then $\ddot{y}_1(t) = -\ddot{x}_2(-t) = -F_2(x(-t))$, and $F_1(y(t)) = -F_2(x(-t))$ because $y_2 - y_1 = -(x_1 - x_2)$ and $y_3 - y_1 = -(x_3 - x_2)$ at time $-t$; bodies 2 and 3 are alike. Each velocity changes sign twice, so the map sends $\dot{x}_1(0)$ to $\dot{x}_2(0) = \dot{x}_1(0)$ and $\dot{x}_3(0)$ to itself, and it sends $x_1(0)$ to $-x_2(0) = x_1(0)$ and $x_3(0) = 0$ to itself; it therefore fixes the initial data. By uniqueness the solution equals its image. Body 3's path $z_3(t) = \sum_k \hat{z}_k e^{2\pi i k t/T}$, with $\hat{z}_k = u^{2\nu k} \hat{q}_k$, is odd, so $\hat{z}_{-k} = -\hat{z}_k$ and $|\hat{q}_{-k}| = |\hat{z}_{-k}| = |\hat{z}_k| = |\hat{q}_k|$. In $\mathcal{L} = (6\pi/T) \sum_k k |\hat{q}_k|^2$ the terms $k |\hat{q}_k|^2$ and $-k |\hat{q}_{-k}|^2$ cancel in pairs. $\square$

The body at the zero has a past that is the negative of its future: a pair in time. The map of the proof sends $x_i \times \dot{x}_i$ to $-x_{i'} \times \dot{x}_{i'}$ with i' the partner label, so it reverses $\mathcal{L}$; a solution fixed by it has $\mathcal{L} = 0$, exactly as a record fixed by time reversal has no cycle current.

The Lagrange triangle. Three masses at the vertices of an equilateral triangle rotating rigidly at angular velocity ω form the Lagrange relative equilibrium. With $M = \sum_i m_i$ and $\beta = \sum_{i < j} m_i m_j / M^2$, its nontrivial linear exponents solve $\lambda^4 + \omega^2 \lambda^2 + \frac{27}{4} \beta \omega^4 = 0$, so $\lambda^2 = \omega^2 (-1 \pm \sqrt{1 - 27\beta})/2$. The motion is linearly stable exactly when these are real, negative and distinct, that is when $27 \sum_{i < j} m_i m_j < M^2$ (Gascheau [9], Routh [23]). At the boundary $27\beta = 1$ the two imaginary pairs meet at $\pm i\omega/\sqrt{2}$, and beyond it they leave the imaginary axis as a quadruple $\{\lambda, -\lambda, \bar{\lambda}, -\bar{\lambda}\}$. For a star, a planet of mass fraction μ and a test body, $\beta = \mu(1 - \mu)$ and the criterion reads $\mu < \mu_G = (1 - \sqrt{23/27})/2 = 0.0385209$; this is the stability of the triangular points L_4, L_5 , satisfied by all four giant planets, whose Trojans are treated in [28]. For equal masses, $27 \sum_{i < j} m_i m_j = 81 > 9 = M^2$ and $\beta = \frac{1}{3}$, so $\lambda^2 = \omega^2 (-1 \pm i\sqrt{8})/2$ and $\lambda = \omega(\pm 1/\sqrt{2} \pm i)$, since $(1/\sqrt{2} + i)^2 = -\frac{1}{2} + i\sqrt{2}$. The growth rate is

$$\max \operatorname{Re} \lambda = \omega/\sqrt{2}. \quad (1)$$

We derive this directly for unit masses, which proves Theorem 1.4(d).

Proof of Theorem 1.4(d). In complex coordinates rotating at ω , $z_j = e^{i\omega t} w_j$, Newton's equations read $\ddot{w}_j + 2i\omega \dot{w}_j - \omega^2 w_j = \sum_{k \neq j} f(w_k - w_j)$ with $f(v) = v/|v|^3$. The triangle $w_j = Ru^j$ of side $a = \sqrt{3}R$ is an equilibrium: $\sum_{k \neq j} (w_k - w_j)/a^3 = -3w_j/a^3$, so $\omega^2 = 3/a^3$. Since $\partial f/\partial v = -\frac{1}{2}|v|^{-3}$ and $\partial f/\partial \bar{v} = -\frac{3}{2}\hat{v}^2|v|^{-3}$ with $\hat{v} = v/|v|$, a perturbation $w_j = Ru^j + \chi_j$ obeys

$$\ddot{\chi}_j + 2i\omega \dot{\chi}_j - \omega^2 \chi_j = -\frac{1}{2a^3} \sum_{k \neq j} (\chi_k - \chi_j) - \frac{3}{2a^3} \sum_{k \neq j} \hat{v}_{jk}^2 (\bar{\chi}_k - \bar{\chi}_j),$$

where $\hat{v}_{j,j\pm 1}^2 = u^{2j} e^{\mp i\pi/3}$. For $\chi_j = u^{2j} b(t)$ the first sum is $-3u^{2j} b$, and the second is $u^{3j} [e^{-i\pi/3}(u - 1) + e^{i\pi/3}(u^{-1} - 1)] \bar{b} = (i\sqrt{3} - i\sqrt{3}) \bar{b} = 0$. So this mode decouples and $\ddot{b} + 2i\omega \dot{b} - \frac{3}{2}\omega^2 b = 0$; with $b = e^{\lambda t}$, $\lambda^2 + 2i\omega \lambda - \frac{3}{2}\omega^2 = 0$, $\lambda = \omega(-i \pm 1/\sqrt{2})$, and the real form of this complex equation adds the conjugates: $\lambda = \omega(\pm 1/\sqrt{2} \pm i)$, the roots of the quartic at $\beta = \frac{1}{3}$. The remaining modes, χ_j the same for all j (translations) and $\chi_j = u^j b_1(t)$ (rotation and dilation of the triangle), have only the exponents 0 and $\pm i\omega$. $\square$

The eigenvalues of the Jacobian of the equations of motion in the rotating frame, formed by central differences with step 10^{-6} , give $\max \operatorname{Re} \lambda/\omega = 0.707106781$.

7 The figure-eight: one curve, thirds of the period, six triads

All orbits are integrated in double precision with the explicit Runge–Kutta method of order 8 of Dormand and Prince with adaptive step size and continuous (dense) output, DOP853 [12], applied to $\ddot{x}_i = \sum_{j \neq i} m_j (x_j - x_i)/|x_j - x_i|^3$, here with relative and absolute tolerance 10^{-13} . For the published data $\mathcal{L}(0) = x_1 \times \dot{x}_1 + x_2 \times \dot{x}_2 = x_1 \times \dot{x}_1 - x_1 \times \dot{x}_1 = 0$ exactly, and, since $\dot{x}_1 = \dot{x}_2 = -\dot{x}_3/2$ and the mutual distances are $2|x_1|, |x_1|, |x_1|$, the energy is $E = \frac{3}{4}|\dot{x}_3|^2 - 5/(2|x_1|) = -1.287141991766$.

Period. Minimising the distance $|\text{state}(t) - \text{state}(0)|$ in $\mathbb{R}^{12}$ (positions and velocities) over $t \in [6.2, 6.45]$, using the dense output and bounded Brent minimisation, gives $T = 6.3259140121$ with return residual 1.6×10^{-9} . This agrees with the published 6.32591398 to 3.2×10^{-8} , the rounding level of eight-digit data; at tolerance 10^{-11} , T moves by 4.6×10^{-10} .

Choreography. With the labels of the published data, the three bodies follow one curve with

$$x_{i+1}(t) = x_i(t - T/3) = x_i(t + 2T/3) \quad (\text{indices mod } 3),$$

to 1.3×10^{-8} (maximum over 2001 equally spaced $t \in [0, T]$ and over i), which is Theorem 1.4(b) with $\nu = -1$.

Refined data. Keeping the positions and $\dot{x}_1 = \dot{x}_2 = -\dot{x}_3/2$, and solving for $\dot{x}_3$ and T by Levenberg–Marquardt least squares on the twelve components of $\text{state}(T) - \text{state}(0)$, gives $\dot{x}_3 = (-0.932407368139, -0.864731461835)$, $T = 6.325914009781$, residual 9.6×10^{-13} . The change in $\dot{x}_3$ is 2.6×10^{-9} , so the published digits are right to their last place; the choreography then holds to 4.1×10^{-12} .

Six triads per period. The signed area $\Delta(t) = \frac{1}{2}(x_2 - x_1) \times (x_3 - x_1)$ of the triangle of the bodies, sampled at 6001 points of $(-T/12, T - T/12]$ with its sign changes refined by bracketing root-finding, vanishes exactly six times per period. The zeros lie at $kT/6$ to 5.5×10^{-9} . At each one the middle body lies at the midpoint to 4.8×10^{-9} and at the centre of mass to 3.2×10^{-9} , as Theorem 1.4(a) requires. The middle bodies are 3, 2, 1, 3, 2, 1 in turn: each body holds the zero twice per period. The half-length $|s|$ is the same at all six to 1.7×10^{-9} ; at $t = 0$ it is $|x_1(0)| = (0.97000436^2 + 0.24308753^2)^{1/2} = 1.0000000028$. The axis alternates between 165.9312° (at $t = 0$, $180^\circ - \arctan(0.24308753/0.97000436)$) and 14.0688° : as point sets there are only two triads, $\{-s, 0, s\}$ and its mirror image, each visited three times with the roles passed round the bodies (Fig. 2a).

Accuracy of the integration. Sampling every $T/64$, over one period the relative energy error is at most 6.9×10^{-13} and $|\mathcal{L}| \leq 1.7 \times 10^{-12}$. Over 50 periods the bounds are 2.7×10^{-11} and 2.4×10^{-12} .

Harmonics. Over one refined period we take the discrete Fourier coefficients $\hat{q}_k = \frac{1}{4096} \sum_{n=0}^{4095} z_1(nT/4096) e^{-2\pi i kn/4096}$ of $z_1 = x_1 + iy_1$. Those with $3 \mid k$, $|k| \leq 60$, are at most 7.2×10^{-13} of the largest (Theorem 1.4(b)). The largest are $|\hat{q}_{\pm 1}| = 0.5504$, $|\hat{q}_{\pm 2}| = 0.1694$,

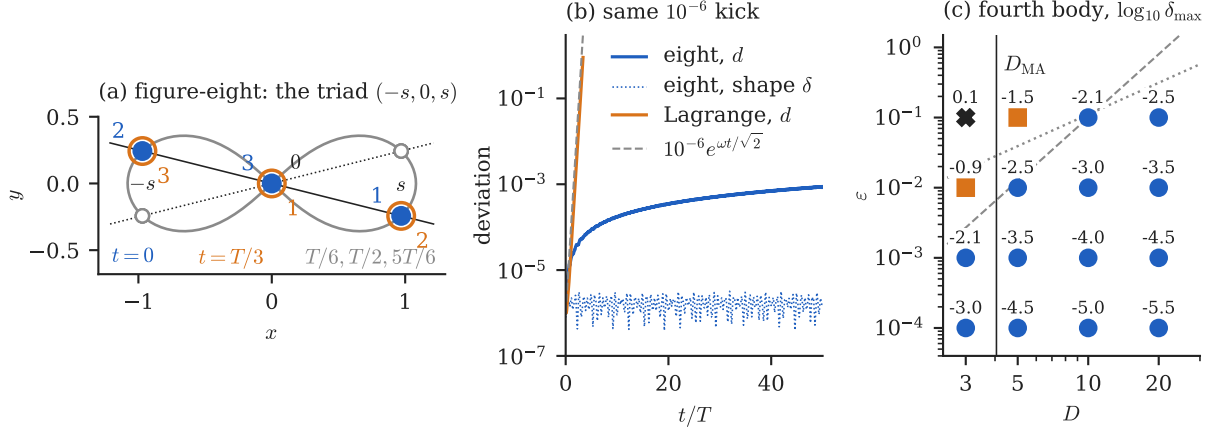


Figure 2: (a) The eight (grey). At $t = 0$ (blue, labels 1, 2, 3) the bodies are at $s, -s, 0$; at $t = T/3$ (orange rings) they occupy the same points with body 1 at the zero. Open circles: the mirror triad at $T/6, T/2, 5T/6$. (b) Deviation after the same 10^{-6} kick: for the eight, the equal-time distance d_{eq} and the shape deviation δ ; for the Lagrange triangle, d_{eq} (stopped at 1), with the rate (1) dashed. (c) $\log_{10} \delta_{\max}$ over $100T$ for a fourth body of mass ε at distance D : kept (blue), deformed (orange), broken (black; value before break-up). Dashed: $\varepsilon/D^3 = 10^{-4}$; dotted: $\delta_{\max} = 10^{-2}$ from (2); vertical: the Mardling–Aarseth distance D_{MA} .

$|\hat{q}_{\pm 4}| = 0.02798$ and $|\hat{q}_{\pm 5}| = 0.01270$; even harmonics are present. On the computed orbit $\max_t |x_3(-t) + x_3(t)| = 2.9 \times 10^{-12}$ and $\max_k ||\hat{q}_k| - |\hat{q}_{-k}|| / \max |\hat{q}| = 3.8 \times 10^{-13}$, as Theorem 1.4(c) requires; the power $\sum |\hat{q}_k|^2$ is 0.332616 in $k > 0$ and in $k < 0$, and $(6\pi/T) \sum k |\hat{q}_k|^2 = 5.7 \times 10^{-13}$: the eight is a sum of balanced counter-rotating pairs of circles.

8 A triad in motion against a triangle at rest

The Lagrange triangle gets the same period, $\omega = 2\pi/T = 0.9932454496$, with side $a = (3/\omega^2)^{1/3} = 1.4487808456$ from $\omega^2 = 3/a^3$ (Section 6), and both orbits the same kick, $+10^{-6}$ to x of body 1. $d_{\text{eq}}(t)$ is the equal-time distance in $\mathbb{R}^6$ to the unperturbed orbit (for Lagrange, the exact rotation), sampled every $T/64$ on $[0, 50T]$; $\delta(t)$ is the shape deviation of Section 9. The Lagrange integration stops when d_{eq} reaches 1.

The figure-eight’s d_{eq} grows linearly: 1.70×10^{-5} at T , 1.72×10^{-4} at $10T$ and 8.57×10^{-4} at $50T$. Its maximum is 8.9×10^{-4} , and the least-squares slope of $\log \max_{r \leq t} d_{\text{eq}}(r)$ against $\log t$ over $[5T, 50T]$ is 1.003 . This is drift along the family’s neutral directions (phase, orientation); its shape stays within 3.2×10^{-6} of the eight. The Lagrange d_{eq} passes 0.1 at $2.875T$ and 1 at $3.389T$, and its shape deviation reaches 0.87 . Its rate, the least-squares slope of $\log d_{\text{eq}}$ against t from the first time $d_{\text{eq}} \geq 10^{-5}$ to the first time $d_{\text{eq}} > 10^{-2}$, is 0.702194 , against $\omega/\sqrt{2} = 0.702331$, a difference of 0.02% .

We repeated the comparison for eight random kicks of norm 10^{-6} , with independent standard normal components in $\mathbb{R}^6$, the centre-of-mass part removed and the result rescaled to norm 10^{-6} . The Lagrange d_{eq} passes 0.1 after every one of them, with fitted rates 0.63197 – 0.70224 . The figure-eight’s shape stays within 5.3×10^{-6} , and its equal-time distance reaches at most 1.68×10^{-3} ; the growth is phase drift, linear in time and in the energy change.

D	10^{-4}	$\delta_{\max}$ over 100 T for $\varepsilon =$ 10^{-3}	10^{-2}	10^{-1}	outer orbit (P_{out}/T ; fate)
3	9.26×10^{-4}	7.45×10^{-3}	0.132^d	1.21^b	2.980; expelled in all four
5	3.27×10^{-5}	3.26×10^{-4}	3.20×10^{-3}	0.0315^d	6.411; bound, $r/D \in [0.893, 1]$
10	9.30×10^{-6}	9.29×10^{-5}	9.22×10^{-4}	8.33×10^{-3}	18.134; bound
20	3.19×10^{-6}	3.18×10^{-5}	3.145×10^{-4}	2.87×10^{-3}	51.291; bound

Table 2: Stability map of the triad with a fourth body; d deformed, b broken (at $15.43 T$), others kept. $P_{\text{out}} = 2\pi\sqrt{D^3/3}$. Control $\varepsilon = 0$: floor 5.2×10^{-8} .

9 A fourth body

Set-up and measure. A body of mass ε starts at distance D from the centre of mass of the refined eight, in direction $+x$, with circular two-body relative speed $\sqrt{(3+\varepsilon)/D}$ in direction $+y$, in the barycentric frame (the triad is shifted by $-\varepsilon/(3+\varepsilon)$ and the fourth body by $3/(3+\varepsilon)$ of the relative position and velocity), so that $3X_{\text{triad}} + \varepsilon x_4 = 0$. The grid is $\varepsilon \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$, $D \in \{3, 5, 10, 20\}$, with integrations over $100 T$ at tolerance 10^{-12} , sampled every $T/64$. With $Y(t)$ the triad positions about the triad's own centre of mass and $Y_8(\tau)$ the same for one period of the eight,

$$\delta(t) = \min_{\tau \in [0, T], \theta} \|Y(t) - R_\theta Y_8(\tau)\|_F,$$

with R_θ the rotation by θ . For fixed τ the minimum over rotations is closed-form: $\min_\theta \|Y - R_\theta Z\|_F^2 = \|Y\|_F^2 + \|Z\|_F^2 - 2(\langle Y, Z \rangle^2 + (\sum_i Z_i \times Y_i)^2)^{1/2}$, because $\langle Y, R_\theta Z \rangle = \cos \theta \langle Y, Z \rangle + \sin \theta \sum_i Z_i \times Y_i$. Y_8 is interpolated by a periodic cubic spline through 4096 samples of one period, and τ is found on a 2048-point grid followed by golden-section refinement (a 20001-point scan with Brent refinement gives the same δ to 3.6×10^{-10}). A case is *kept* if all inner distances stay below 3 and $\delta_{\max} < 10^{-2}$, *deformed* if they stay below 3 but $\delta_{\max} \geq 10^{-2}$, and *broken* otherwise (the integration stops when an inner distance reaches 3). We write $\eta = \varepsilon/D^3$ for the tidal parameter.

The map (Table 2, Fig. 2c). All 15 cases with $\eta < 10^{-3}$ survive. At fixed D , $\delta_{\max}$ is linear in ε : among kept cases with $\varepsilon \leq 10^{-2}$, $\delta_{\max}/\varepsilon$ varies by at most a factor 1.244 at $D = 3$ and 1.021 for $D \geq 5$. In η the grid splits cleanly: every case with $\eta \leq 10^{-4}$ is kept and every case with $\eta \geq 3.7 \times 10^{-4}$ is not, so the threshold lies in $(10^{-4}, 3.7 \times 10^{-4})$: $(D, \varepsilon) = (5, 0.1)$ and $(3, 0.01)$ are deformed and $(3, 0.1)$ breaks. Over the kept cases $\delta_{\max}/\eta$ ranges from 40.0 to 254.8, growing with D for $D \geq 5$: the tidal parameter alone does not set the deformation. The next paragraph identifies what does.

What sets it: the spin the fourth body lends and takes back. Let $Y_i = (\xi_i, \zeta_i)$ be the triad positions about its centre of mass ($\sum_i Y_i = 0$, unit masses) and let the fourth body be at distance D in direction $\hat{R} = (\cos \psi, \sin \psi)$, $\psi = \Omega t$, $\Omega = \sqrt{(3+\varepsilon)/D^3}$. Expanding the potential energy $-\sum_i \varepsilon/|D\hat{R} - Y_i|$ to second order in $|Y_i|/D$, the monopole term drives the outer orbit, the dipole term cancels because $\sum_i Y_i = 0$, and the quadrupole term is

$$V = -\frac{\varepsilon}{2D^3} \sum_i [3(\hat{R} \cdot Y_i)^2 - |Y_i|^2] = -\frac{3\varepsilon}{2D^3} \left[\frac{1}{2} \Delta I \cos 2\psi + I_{xy} \sin 2\psi \right] + \text{const},$$

with $I_{xx} = \sum \xi_i^2$, $I_{yy} = \sum \zeta_i^2$, $I_{xy} = \sum \xi_i \zeta_i$ and $\Delta I = I_{xx} - I_{yy}$, the constant not depending on ψ . Rotating the triad by ϕ changes ψ to $\psi - \phi$, so the torque on it is $\Gamma = -\partial V/\partial \phi = \partial V/\partial \psi = (3\varepsilon/2D^3)(\Delta I \sin 2\psi - 2I_{xy} \cos 2\psi)$. Averaged over the inner period, with $\langle I_{xy} \rangle = 3.8 \times 10^{-9}$ and $\langle \Delta I \rangle = 1.641896$ (means over 4096 equally spaced samples of one period of the refined eight,

which is elongated along x), this is $\Gamma = (3\varepsilon/2D^3)\langle\Delta I\rangle \sin 2\Omega t$. Integrating from $\mathcal{L}_{\text{in}}(0) = 0$ gives the triad a spin

$$\mathcal{L}_{\text{in}}(t) = \frac{3\varepsilon\langle\Delta I\rangle}{4\Omega D^3} (1 - \cos 2\Omega t), \quad \max_t \mathcal{L}_{\text{in}} = \frac{3\varepsilon\langle\Delta I\rangle}{2\Omega D^3}.$$

The torque is a pair: it is odd under $t \mapsto \pi/\Omega - t$, so over each half outer orbit its positive and negative halves cancel, and the spin returns to zero. In integrations at $\varepsilon = 10^{-3}$, $D \in \{5, 7, 10, 14, 20, 28\}$, the measured $\max |\mathcal{L}_{\text{in}}|$, with $\mathcal{L}_{\text{in}} = \sum_i Y_i \times \dot{Y}_i$ about the triad's centre of mass, exceeds this maximum by the ratios 1.1361 ($D = 5$), 1.0244 ($D = 10$) and 1.0030 ($D = 28$); the measured ratio $\delta_{\text{max}}/\max |\mathcal{L}_{\text{in}}|$ tends to 1.985; and the least-squares slopes against $\log D$ for $D \geq 10$ are -1.535 ($\log \delta_{\text{max}}$) and -1.520 ($\log \max |\mathcal{L}_{\text{in}}|$), against $-\frac{3}{2}$ from the formula. Combining the measured $\delta_{\text{max}} \simeq 2 \max |\mathcal{L}_{\text{in}}|$ with $\max \mathcal{L}_{\text{in}}$ and $\Omega \simeq \sqrt{3}D^{-3/2}$ for $\varepsilon \ll 3$ gives, at leading order,

$$\delta_{\text{max}} \simeq 2 \max |\mathcal{L}_{\text{in}}| \simeq \sqrt{3} \langle\Delta I\rangle \varepsilon D^{-3/2} = 2.8438 \varepsilon D^{-3/2}. \quad (2)$$

Across the grid of Table 2, δ_{max} equals (2) times 1.24–1.28 ($D = 5$), 0.93–1.03 ($D = 10$) and 0.90–1.00 ($D = 20$). The governing parameter is $\varepsilon D^{-3/2} \propto (\varepsilon/D^3) P_{\text{out}}$, the tidal strength times the time over which it acts coherently. By (2), $\delta_{\text{max}} = 10^{-2}$ at $\varepsilon D^{-3/2} = 10^{-2}/2.8438 = 3.516 \times 10^{-3}$, which in terms of $\eta = (\varepsilon D^{-3/2})D^{-3/2}$ is 3.15×10^{-4} at $D = 5$, 1.11×10^{-4} at $D = 10$ and 3.93×10^{-5} at $D = 20$. The deformation is reversible: at $D = 20$, $\varepsilon = 10^{-3}$, where half an outer period is $\pi/\Omega = 25.641 T$, the minimum of δ within $1.5 T$ of each of the first three half outer periods is 5.96×10^{-8} , 4.22×10^{-8} and 6.66×10^{-8} , the $\varepsilon = 0$ floor, against $\delta_{\text{max}} = 3.18 \times 10^{-5}$. At tolerance 10^{-13} , δ_{max} is 3.150319×10^{-2} at (5, 0.1) (unchanged) and 0.1325148 at (3, 0.01) (against 0.1324405 at 10^{-12}); the chaotic case (3, 0.1) breaks at $10.71 T$ (against $15.43 T$).

The new pair and the Mardling–Aarseth boundary. For a circular coplanar prograde outer orbit, Mardling and Aarseth's stability criterion for hierarchical triples [17] is $D > 2.8 a_{\text{in}}(1 + q_{\text{out}})^{2/5}$, $q_{\text{out}} = \varepsilon/3$ the outer-to-inner mass ratio. With $a_{\text{in}} = (3(T/2\pi)^2)^{1/3} = 1.448781$ (by Kepler's third law, the semi-major axis of a binary of mass 3 and period T , equal to the side a of the Lagrange triangle of Section 8), $D_{\text{MA}} = 4.0566$ – 4.1101 over the four values of ε . At $D = 3$, inside the boundary, the fourth body is expelled in every case, ending at $r/D = 83.2$, 47.5, 401.7, 85.1 with positive two-body energy. At $D \geq 5$ the 3 + 1 pair stays bound, though not exactly circular: at $D = 5$, $r/D \in [0.893, 1.000]$, an excursion of up to 0.1065. The same excursion occurs at $\varepsilon = 0$: it is the triad's own quadrupole acting on a point-mass circular start, and it falls to 0.0238–0.0258 at $D = 10$ and 0.0059–0.0062 at $D = 20$. Energy is conserved to 1.1×10^{-9} in the surviving cases and to 5.2×10^{-7} in the broken case, and at tolerance 10^{-13} to 9.8×10^{-11} and 2.7×10^{-9} . The complementary limit, a third body of vanishing mass near one member of a pair, is Hill's problem, whose two throats at L_1 and L_2 are treated in [28].

The fourth element. The figure-eight is the triad in motion, with the zero passing round the bodies; the rigid triangle, with no body at the zero, is unstable. The fourth body and the triad form a new pair about their joint centre, $3X_{\text{triad}} + \varepsilon x_4 = 0$, a triad seen from outside: outside the Mardling–Aarseth boundary it lends the triad a spin and takes it back every half outer orbit; inside it the new pair does not form.

10 Discussion: one involution across records, fields and orbits

Each system of this paper carries an involution ι , a map equal to its own inverse. Every quantity Q splits into an even half $\frac{1}{2}(Q + \iota Q)$ and an odd half $\frac{1}{2}(Q - \iota Q)$, and an odd quantity vanishes

System	Involution ι	Odd quantity	On the fixed set; computed value
Measurement record, §§2, 4	time reversal, $J \mapsto J^\top$	cycle currents $J - J^\top$	zero iff detailed balance; four-outcome instrument: $\max J - J^\top = 0.00970$
Singlet walk, §3	sequence reversal	$\log(P_{\rightarrow}/P_{\leftarrow})$	zero for the stationary record; total $c \operatorname{artanh}(c)$ nats from the seed
Three-phase field, §5	$K(\tau) \mapsto K(\tau)^\top; b \leftrightarrow c$	swept area Φ	zero iff $ I_+ = I_- $; example $\Phi = 6.5578$
Figure-eight, §§6, 7	$(t, x_1, x_2, x_3) \mapsto (-t, -x_2, -x_1, -x_3)$	angular momentum $\mathcal{L}$	zero harmonic by harmonic; computed 5.7×10^{-13}
Lagrange triangle, §§6, 8	$\lambda \mapsto -\bar{\lambda}$ on the spectrum	$\operatorname{Re} \lambda$	zero iff $27 \sum m_i m_j \leq M^2$ (stable iff $<$); equal masses $\pm\omega/\sqrt{2}$, computed 0.702194
Fourth body, §9	$t \mapsto \pi/\Omega - t$	averaged torque Γ	spin returns to zero each half outer orbit; δ returns to the 5×10^{-8} floor

Table 3: One involution, six instances. In each, the odd half vanishes on the set fixed by ι .

on every object that ι fixes. Table 3 lists the six instances; each entry is proved or computed in the section cited.

Three structural points connect the two halves of the paper.

The pair. In the record, the reversible half S and the odd half A are the even and odd parts of C under time reversal, and Theorem 1.1 measures the cost of the odd half: it can take from the entropy rate at most one quarter of what it produces, and for the four-outcome instrument it takes within 0.15% of that quarter. In the field, the forward and backward components $\alpha_{\pm}$ are exchanged by the same involution, and the swept area is their signed difference. In the figure-eight, the harmonics $\hat{q}_k$ and $\hat{q}_{-k}$ are counter-rotating circles of equal radius, paired by the time-reversal symmetry, and the angular momentum, their signed difference, vanishes.

The triad. The phases $1, u, \bar{u}$ of the three coils, of the three bodies at thirds of the period, and of the vertices of the Lagrange triangle are the same triad of the circle, with $1 + u + \bar{u} = 0$. In the field the identity removes the common mode I_0 ; in the choreography it removes every harmonic divisible by three; in the Lagrange triangle it decouples the mode $\chi_j = u^{2j}b$ that carries the instability. The same identity makes a probe qubit coupled to a maximally mixed spin-1 decohere completely at $\frac{1}{3}$ and $\frac{2}{3}$ of its revival period [27]. Among three equal masses, the triad with a body at the zero, the figure-eight, is linearly stable [21] and keeps its shape to 3.2×10^{-6} after a kick, while the rigid triangle, with no body at the zero, leaves at $\omega/\sqrt{2}$; and a triangle of general masses stands only around a dominant centre: since $\sum_{i<j} m_i m_j = m_1(M - m_1) + m_2 m_3 \geq m_1(M - m_1)$ for the largest mass m_1 , the criterion $27 \sum_{i<j} m_i m_j < M^2$ forces $27x(1 - x) < 1$ with $x = m_1/M$, that is $x < \mu_G$ or $x > 1 - \mu_G$, and $x \geq \frac{1}{3} > \mu_G$ leaves $m_1 > (1 - \mu_G)M = 0.9614791 M$ [9, 23, 28].

The fourth element. The observer writes the record; the stationary singlet record, written from the fourth state of $\frac{1}{2} \otimes \frac{1}{2}$, is reversible, and its whole arrow is the seed correlation. The fourth body, watching the triad from outside, forms a new pair with it, $3X_{\text{triad}} + \varepsilon x_4 = 0$, and its effect on the triad is itself a pair in time: the spin lent over one quarter of the outer orbit is returned over the next.

Outlook. With the raw tomographic counts behind the matrix C of Section 4, the sampling uncertainty of its entropy production, $\text{EP} = 0.00686$ bits per use, can be computed and set

against it. The next step for the fourth body is to carry the quadrupole torque of Section 9 to eccentric and inclined outer orbits.

11 Data and sources

No observational dataset is used. The external inputs are the following.

- *Transition matrix of the processed four-outcome instrument* (Section 4): the 4×4 column-stochastic matrix C displayed there, to eight decimal places, is taken from [26], §9, Eq. (24) (repeated in its App. E, Eq. (50)), together with the maximal positivity factor 0.708418699729814 ([26], §8, Eq. (23)). In [26], C is obtained from processed, publicly available tomographic estimates of a four-outcome Bell-measurement instrument (effects E_a and conditional output states ϱ_a), physically completed by the Kraus family K_{asr} and expressed in a common basis; that source gives no raw counts or covariance and does not identify the underlying measurement further. Only the matrix C is used here; every quantity in Section 4 and Fig. 1 is computed from it. The values of π , $\text{spec } C$, $H(X_t)$, h , $I(X_t; X_{t+1})$, κ and the bias bound given in [26], §§9–11, agree with ours to the rounding of C .
- *Singlet walk* (Section 3): no external data; all numbers follow from the closed forms of Theorem 1.2, with $c = \cos(\pi/n)$, $d = \cos(2\pi/n)$.
- *Random chains* (Section 4): drawn as described there; no external dataset.
- *Rotating field* (Section 5): the currents $(1, 0.8\angle -120^\circ, 1.1\angle 118^\circ)$ are an illustrative choice.
- *Initial data of the figure-eight* (computed by Simó and published by Chenciner and Montgomery [2]): $x_1 = -x_2 = (0.97000436, -0.24308753)$, $x_3 = (0, 0)$, $\dot{x}_3 = -2\dot{x}_1 = -2\dot{x}_2 = (-0.93240737, -0.86473146)$, published period 6.32591398. The refined data of Section 7 are derived from these.
- *Lagrange triangle*: the characteristic quartic and the stability criterion of Gascheau and Routh [9, 23], derived in Section 6 for unit masses.
- *Mardling–Aarseth criterion* [17]: constant 2.8 and exponent $2/5$, used with outer eccentricity 0 and zero mutual inclination.
- *Integration method*: the order-8 Dormand–Prince pair with dense output, DOP853 [12].
- *Stability of the eight* (context, no numbers taken): Roberts [21]; numerical studies by Simó [25]; discovery by Moore [19].
- *Physical constants*: none enter numerically. The three-body problem is in dimensionless units, $G = 1$; record costs are stated in units of $k_B T_0$.

All other parameters (kicks, ε , D , tolerances, sampling steps, grid sizes, numbers of settings) are choices stated where they are used.

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